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# Urinary tract infections in women: classification

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# Definitions

- Uncomplicated urinary tract infection: simple acute cystitis without fever, risk factors, or secondary diseases. Reacts quickly to treatment.
- Complicated urinary tract infection: Infection of the urinary tract (bladder, kidneys) in patients with risk factors or secondary diseases.
- Pyelonephritis: Urinary tract infection with kidney involvement and corresponding symptoms including fever, kidney pain, feeling ill.
- Asymptomatic bacteriuria: Evidence of ≥ 100’000 bacteria in 1 ml urine without typical symptoms of cystitis.

# Epidemiology

Urinary tract infections are one of the most common reasons for medical consultations. About 50 % of all women experience at least one urinary tract infection in their lives. Annually, 20 to 40 in 1000 general practice consultations are for urinary tract infections. 20 % of all antibiotics used in clinical practice are prescribed for the treatment of UTIs. The first frequency peak occurs following first sexual contact, with further peaks upon pregnancy and then within menopause.

Risk factors for urinary tract infections are as following:

- Previous urinary tract infection
- Sexual intercourse
- Use of diaphragms and spermicides
- Urinary tract infections during childhood (below 15 years)
- Mother with urinary tract infections
- Use of tampons and condoms
- Use of antibiotics in the preceding 2 to 4 weeks
- Genetic factors
- Advanced age

# Diagnostics

The medical history should include the frequency of urinary tract infections and risk factors (see above). In addition, complicated urinary tract infections must be distinguished from uncomplicated ones. Secondary and underlying diseases should therefore be searched for. Asking about fever, kidney pain and a general feeling of illness likely provide information about pyelonephritis. A physical examination is usually not necessary with uncomplicated urinary tract infections, if a reliable medical history has been conducted.

## Urine tests

Test strips are cheap and generally suffice for diagnosing an uncomplicated urinary tract infection. Positive nitrite test: Nitrites are released by E. Coli bacteria, and thus have a high predictive value for E. Coli infections (approx. 90 %). However, as other microorganisms can be present, the test's sensitivity is only approx. 50 %. Proof of leukocytes by means of leukocyte esterase is also very sensitive (approx. 90 %). If both tests are negative, the probability of a urinary tract infection proves to be low. Urine sediment testing and urine cultures are usually not necessary for uncomplicated urinary tract infections.

Figure 1: Urine test strip. Test strips are immersed in fresh midstream urine and then analyzed against a reference (with kind permission of Dr. G. Ryu).

## Urine sediment

Urine sediment can be used to identify leukocyturia, hematuria, and bacteiuria. Owing to the quick availability and quantitative information, such as bacteria counts, a rapid diagnosis can be made to supplement the test strip. In addition, with persistent symptoms, the bacteria count can help predict treatment response. Overall, however, urine sediment testing is of little value.

## Urine culture

Urine cultures are carried out in the following situations:

- Unsuccessfully treated primary uncomplicated urinary tract infection
- Recurrence of urinary tract infection
- Complicated urinary tract infection
- Abnormal test strip upon pregnancy
- Suspected pyelonephritis

## Gynecological examination

A gynecological examination is rarely necessary with uncomplicated urinary tract infections. For recurring, persistent, or complicated urinary tract infections, a gynecological examination with vaginal examination and swabs of the vaginal excretions and urethra is recommended. The examination should look for signs of genital atrophy and indications of downward displacement, in particular cystocele, prolapse of the uterus, or vaginal apex, along with any residual urine (ultrasound).

For complicated urinary tract infections, a urine culture should be obtained to monitor the success of the treatment. In such cases, treatment duration should exceed 3 days, with experts recommending 10 to 13 days (cf. G02H).

# Pathogenesis of urinary tract infections

Cystitis most commonly occurs due to the ascension of perineal, perianal, and intravaginal bacteria (figure 2). Bacteria can travel trans-urethrally into the bladder through mechanical manipulation, such as sexual intercourse, catheterization, or water flows from shower toilets. Hematogenic and lymphogenic infection pathways are additionally possible, particularly with kidney infections, for example with staphylococcus or tuberculosis bacteria.

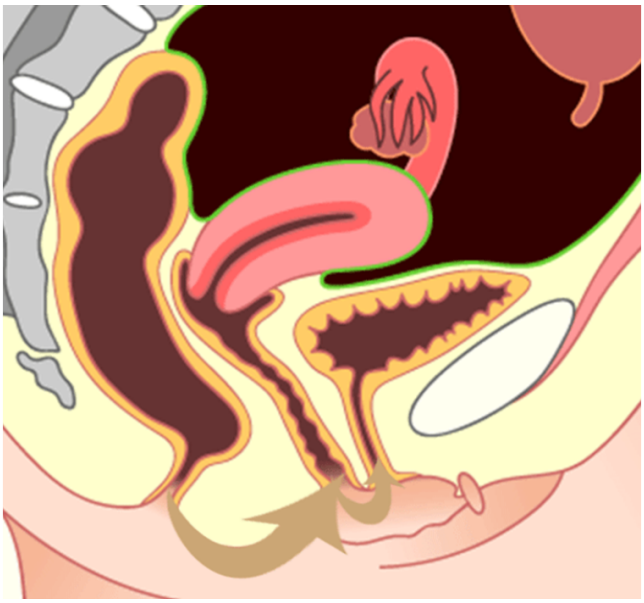

Figure 2: Intestinal microorganisms colonize the perianal and vulvar skin, as well as the vagina, and ascend trans-urethrally from there into the bladder.

Practical examples of complicated urinary tract infections. If the following conditions are met, the urinary tract infection is classified as complicated:

- Bladder catheter (transurethral and suprapubic catheter)
- Functional abnormality like pelvic organ prolapse, post-void residual volume of (> 100 ml)
- Diabetes mellitus
- Functional or anatomical urinary tract disorders
- Immunosuppression
- Infection in children
- Infection in men
- Voidance disorders
- Kidney failure
- Pregnancy
- Urolithiasis

- Previous urinary tract surgery
- Polycystic kidney disease
- History of kidney transplantation

# Predisposing factors

If factors explaining the definition of a complicated urinary tract infection are not present, the following situations may lead to urinary tract infections and recurring urinary tract infections:

Patient-related:

- Immune system defects in the bladder with lacking antibodies, for example immunoglobulin A.
- Uro-mucosal disorders with bladder mucosa defects, which in turn facilitate bacterial adhesion.
- Postmenopausal changes to vaginal secretions; a lack of lactobacilli leading to an increase in pH, thus providing the perfect environment for uro-pathogenic bacteria.

Bacteria-related:

- Virulence factors: Fimbriae are a virulence factor and potentially facilitate bacterial adhesion (E. Coli fimbriae).

# Bacterial spectrum of urinary tract infections

Approximately 80 % of community-acquired urinary tract infections are caused by E. Coli, compared with only 40 to 50 % of hospital-acquired infections. The frequency of other pathogens such as Klebsiellae, Pseudomonas aeruginosa, Enterococcus faecalis, Streptococci, Staphylococcus epidermidis, Proteus mirabilis etc. increases under hospital conditions. These pathogens exhibit stronger resistance to antibiotics.

Table 1: Urinary tract infection bacterial spectrum

| Community-acquired | Hospital-acquired |
|---|---|
| E. coli (~ 80 %) | E. coli (~ 50 %) |
| Staph. Saprophyticus | Klebsiellae |
| Proteus mirabilis | Pseudomonas aeruginosa |
| Enterococcus faecalis | Enterococcus faecalis |
| Pseudomonas aeruginosa | Streptococci |
| Klebsiellae | Staph. epidermidis |
| | Proteus mirabilis |

A special culture medium is required to detect certain microorganisms. These challenging microorganisms are listed in table 2.

Challenging microorganisms that require a special medium or detection method:

- Anaerobes
- Gardnerella vaginalis
- Neisseria gonorrhoeae
- Chlamydia trachomatis
- Mycoplasma hominis
- Ureaplasma urealyticum
- Herpesviruses
- Candida/Fungi
- Trichomonas vaginalis

Special culture media (cf. G03B) must therefore be requested if a urinary tract infection is very likely, whereas urine culture has provided no indication of its cause.

# Further diagnostics

Invasive and imaging diagnostics are unnecessary with simple urinary tract infections. Structural and functional anomalies are rare. Exceptions are previous urogenital tract interventions and persistent hematuria. In such cases, ultrasound of the urinary tract, including the kidneys, and cystoscopy are recommended.

# Literature


| Nr. | Title | Author | Journal | Volume | Pages | Year |
|---|---|---|---|---|---|---|
| [B01] | Management of urinary tract infection in general practice: a cost-effectiveness analysis | Fenwick EA, Briggs AH, Hawke CI | Br J Gen Pract | 50(457) | 635-639 | 2000 |
| [B02] | Risk factors for recurrent urinary tract infection in young women | Scholes D, Hooton TM, Roberts PL, Stapleton AE, Gupta K, Stamm WE | J Infect Dis | 182(4) | 1177-1182 | 2000 |
| [B03] | Cranberries for preventing urinary tract infections | Jepson RG, Craig JC | Cochrane Database Syst Rev | CD001321 | | 2008 |
| [B04] | Asymptomatic bacteriuria and symptomatic urinary tract infections in pregnancy | Schnarr J, Smaill F | Eur J Clin Invest | 38 | 50-57 | 2008 |
| [B05] | Guidelines for the diagnosis and management of recurrent urinary tract infection in women | Dason S, Dason JT, Kapoor A | Can Urol Assoc J | 5(5) | 316-322 | 2011 |

| Nr. | Title | Author | Journal | Volume | Pages | Year |
|---|---|---|---|---|---|---|
| [B06] | Clinical practice. Uncomplicated urinary tract infection | Hooton TM | N Engl J Med | 366(11) | 1028-1037 | 2011 |
| [B07] | Urinary tract infection | Gupta K, Trautner B | Ann Intern Med | 156(5) | ITC31 | 2011 |
| [B08] | Incidence, severity, help seeking, and management of uncomplicated urinary tract infection: a population-based survey | Butler CC, Hawking MK, Quigley A, McNulty CA | Br J Gen Pract. | 65(639) | e702-707 | 2015 |

| Nr. | Title | Author | Journal | Volume | Pages | Year |
|---|---|---|---|---|---|---|
| [P01] | Prevention and treatment of uncomplicated lower urinary tract infections in the era of increasing antimicrobial resistance-non-antibiotic approaches: a systemic review | Wawrysiuk S, Naber K, Rechberger T, Miotla P | Arch Gynecol Obstet | 300(4) | 821-828 | 2019 |

| Nr. | Title | Author | Journal | Volume | Pages | Year |
|---|---|---|---|---|---|---|
| [P02] | Contemporary management of uncomplicated urinary tract infections | Guay DR | Drugs | 68(9) | 1169-1205 | 2008 |
| [P03] | Immunostimulation with Escherichia coli extract: prevention of recurrent urinary tract infections | Ha US, Cho YH | Int J Antimicrob Agents | 31 | 63-67 | 2008 |
| [P04] | Probiotics for prevention of recurrent urinary tract infections in women: a review of the evidence from microbiological and clinical studies | Falagas ME, Betsi GI, Tokas T, Athanasiou S | Drugs | 66 | 1253-1261 | 2006 |
| [P05] | trial of cranberry-lingonberry juice and Lactobacillus GG drink for the prevention of urinary tract infections in women | Kontiokari T, Sundqvist K, Nuutinen M, Pokka T, Koskela M, Uhari M | BMJ | 322 | 1571 | 2001 |
| [P06] | Response to antibiotics of women with symptoms of urinary tract infection but negative dipstick urine test results: double blind randomised controlled trial | Richards D, Toop L, Chambers S, Fletcher L | Bmj | 331 | 143 | 2005 |

| Nr. | Title | Author | Journal | Volume | Pages | Year |
|---|---|---|---|---|---|---|
| [P07] | Urinary tract infections during pregnancy | Delzell JE, Lefevre ML | Am Fam Physician | 61 | 713-721 | 2000 |